FITBUDDY

AI Fitness Plan Generator using Gemini Models

Solution Requirements

Date	01 October 2026
Team ID	SWTID-2026-2430
Project Name	Team Enrollment Confirmation – FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	4 Marks
Team Members	Divya.D, Deepika.M, Dharani.T, Divya shree.R, DivyaDharshini.V

Define Functional and Non-Functional Requirements

This document describes what FitBuddy will do and the quality criteria it should meet. Functional requirements specify system features; non-functional requirements specify how well those features should work.

Project scope: FitBuddy accepts basic user details and fitness goals, generates a personalized 7-day workout plan using Google Gemini models, provides a concise nutrition or recovery tip, supports feedback-based plan updates, stores plan history in SQLite, and provides an admin view.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can submit the fitness form. If account login is included, credentials must be validated securely; otherwise, the initial version may allow a basic form without accounts.	Medium
2	Authorization levels	Admin view is intended for oversight of stored plans and records. Admin-only access should be enforced if the view exposes user or plan data.	Medium
3	External interfaces	The FastAPI backend connects to Google Gemini models through the configured API. HTML/CSS/Jinja2 pages provide the user interface; API documentation is available for testing.	High
4	Transactions processing	Validate user inputs, send a structured request to Gemini, receive and display a 7-day plan and tip, and process feedback to generate an updated plan. Save original and updated plans to SQLite.	High
5	Reporting	Display the generated weekly workout plan and nutrition/recovery tip. The admin view should support oversight of saved user details, feedback, and plan history.	Medium
6	Business rules, etc.	Plans should reflect submitted goals and preferences, contain seven days of structured workouts, include a concise nutrition or recovery tip, and support revisions based on user feedback.	High
7	Compliance to laws or regulations	Collect only information needed for plan generation; explain how user inputs and plan history are used. Follow applicable privacy requirements and protect stored data and API credentials.	High
8	Other	Show clear validation messages for missing or invalid inputs, and provide a useful error message if AI generation fails. Present fitness suggestions as general information.	Medium

FITBUDDY — Solution Requirements (continued)

Step 2: Non-Functional Requirements (NFR)

These target metrics are proposed acceptance criteria for the project and can be adjusted after testing on the actual development setup.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Pages and saved records should load promptly; generation time depends partly on the Gemini service and network.	Local pages load within 2 seconds under normal test conditions; show a progress state while generating.
2	Scalability	The application should support growth in users and plan records without changing its core workflow.	Pass functional tests with at least 20 sequential test requests; document any concurrency limits of the local setup.
3	Security & Data Privacy	Protect API keys, validate inputs, avoid exposing sensitive details in error messages, and restrict admin access where user data is shown.	API key stored outside source code; server-side validation enabled; admin route access checked.
4	Reliability & Availability	Handle AI/API errors, invalid responses, and database failures gracefully without losing existing saved records.	Show a clear error for failed generation; confirm saved history remains intact after restart.
5	Usability & Accessibility	Forms and results should be readable, easy to navigate, and usable on common desktop and mobile screen sizes.	All main tasks can be completed using labeled form fields and visible feedback; test at mobile and desktop widths.
6	Other — Maintainability & Portability	Keep routes, AI prompt logic, templates, and database operations organized so the project can be maintained and run locally.	Project starts using documented setup steps; dependencies and required environment variables are documented.

Team Work Allocation

Responsibilities are assigned in the same order as the team-member list provided.

S.No	Team Member	Assigned Responsibility	Expected Contribution
1	Divya.D	Requirements and coordination	Define user inputs, goals, workflows, and requirements; coordinate and consolidate documentation.
2	Deepika.M	Gemini AI integration and prompts	Design prompts for structured 7-day plans, nutrition/recovery tips, and feedback-based revisions.
3	Dharani.T	FastAPI backend and routing	Implement form handling and routes for plan generation and feedback; connect routes to AI functions.
4	Divya shree.R	Frontend and user experience	Build HTML/Jinja2 pages for inputs, plans, tips, and feedback; style responsive layouts.
5	DivyaDharshini.V	Database, testing, deployment support	Support SQLite/SQLAlchemy plan history, test main workflows, and verify local deployment/API docs.

Shared team responsibility: All members review the final output, check that plans are clearly presented, and prepare the project demonstration. Fitness suggestions are general information and should be adapted to individual needs.